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Creating and applying the Ecosystem Pie Model as a design theory

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ABSTRACT

Innovation ecosystems have become pivotal phenomena in a world of increasing specialization and complexity, in which single ventures cannot develop and commercialize technology-driven value propositions alone. In this setting, various firms and other actors need to work together in creating and building an innovation ecosystem around a shared value proposition. The Ecosystem Pie Model (EPM), published in *Long Range Planning* in 2020, is the first comprehensive tool for mapping and assessing the risk profile of a prospective innovation ecosystem. We describe, and reflect on, how the so-called design theory underlying the EPM was developed. This design theory includes its key constructs, design principles, visual instantiation, detailed application guidelines, and assessment of how mutable its applications are. The initial design theory was used to develop more than 240 ecosystem constellations to assess its applicability and mutability. We also outline various applications of EPM design theory by others. Overall, this paper contributes to the entrepreneurship and innovation literature by developing a complete design theory, one that covers the entire spectrum from theoretical underpinnings to detailed application guidelines and an initial body of applications.

1. Introduction

In today's world of increasing complexity and specialization, a single firm often does not have the competences and resources to develop and commercialize a complex value proposition from start to finish (Appleyard & Chesbrough, 2017; Talmar et al., 2020). A firm seeking to commercialize an innovation therefore needs to create and orchestrate an innovation ecosystem for building an ecosystem-wide value proposition (Adner, 2012), which only materializes when the contribution of each of the different actors is effectively delivered and subsequently combined (Adner & Silverman, 2013; Pushpananthan & Elmquist, 2022; Staudenmayer et al., 2005). The theoretical as well as practical value of the innovation ecosystem construct has attracted a growing number of researchers and practitioners (e.g. Oh et al., 2016; Pushpananthan & Elmquist, 2022; Staudenmayer et al., 2005; Randhawa et al., 2021).

In this paper, we conceive of an *innovation ecosystem* as a network where "joint value is created through innovations whose components are (by definition) both complementary and modular" (Baldwin et al., 2024, p. 3), which resonates well with how many others have defined it (e.g. Adner, 2012; Hou & Shi, 2021; Pushpananthan & Elmquist, 2022; Remneland Wikhamn & Styhre, 2023; Thomas & Ritala, 2022). More specifically, we focus on innovation ecosystems that are envisioned and driven by *focal* actors (e.g. new ventures or inventors) which need a set of other actors to provide complementary value offerings to be able to accomplish a collective value proposition (Adner, 2012).

This conceptualization clearly demarcates an innovation ecosystem from a supply chain that is transactional and sequential in nature, with various companies committing their physical and intellectual assets to enable the entire chain to produce and deliver specific products and services—with the output of one firm being another's input (Mentzer et al., 2001). The need to develop an innovation ecosystem for a focal actor is fundamentally different, because it involves the search for a novel value proposition that requires the focal actor's innovation as well as all relevant complements needed to materialize that value offering in the future. That is, its function is not (yet) to produce and supply but to *search and align* across all (envisioned) complementary value offerings and their prospective suppliers (Talmar et al., 2020). The innovation ecosystem's output in terms of a shared value proposition and associated

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products and services is therefore not yet known, with (at least some of) the complementary offerings not yet existing. And its composition in terms of actors as well as their roles and contributions are also not yet specified. Moreover, the focal innovator often has not yet committed to commercializing its innovation with any specific ecosystem constellation, and therefore wants to explore and compare various different ecosystem constellations before choosing the most promising one (Talmar et al., 2020).

Notably, some innovation ecosystems will, over time, develop into (typically multiple) supply chains or be integrated into existing ones. Examples are innovation ecosystems exploring how additive manufacturing technology can be used to create a new generation of medical implants and devices (Peron et al., 2025) or developing a software-hardware solution for demand-side management of residential heating systems (Peltokorpi et al., 2019). Other innovation ecosystems may evolve into ecosystems (without the explicit 'innovation' adjective) characterized by co-evolution, super-modular interdependence and continued generativity (Jacobides et al., 2018), as in the case of integrating blockchain technology into the onboard computing systems in satellites to create a new generation of satellite-based Internet-of-Things applications (Sylos Labini et al., 2024).

Accordingly, the *interdependency* between actors and their prospective complementary offerings is pivotal in any innovation ecosystem searching for an overarching value proposition and the ecosystem configuration needed to accomplish it (Talmar et al., 2020). This interdependency can strongly confine the focal firm, for instance, when a novel product/service cannot be launched because other parties fail to materialize critical complementary products/services in a timely manner (Adner, 2006; Pushpananthan & Elmquist, 2022). But the same interdependency also enables the ecosystem to leverage comparative advantages and network effects from the complementarities of the actors involved (Adner & Feiler, 2019; Remneland Wikhamn & Styhre, 2023).

With the increasingly pivotal role of innovation ecosystems in various industries, the demand for practical models and tools has also been growing, especially those that address the ecosystem interdependencies and associated risks arising from a focal firm's new product development. In response to this demand, we designed the *Ecosystem Pie Model* (EPM), earlier published in Long Range Planning (Talmar et al., 2020). At the time, the EPM was the first practical method for performing a comprehensive *risk-oriented analysis* of how a focal firm navigates the dependencies of its own new product (development process) on the various complements (to be) developed by other actors in its ecosystem. Many other scholars (e.g. Helman, 2020; Peron et al., 2025; Sourmelis et al., 2024; Sylos Labini et al., 2024), public institutions like Flanders Innovation & Entrepreneurship, and consulting companies such as Berenschot (Krebbekx et al., 2021) have subsequently adopted it.

In this paper, we describe and reflect on the design science approach (Dimov et al., 2023; Romme & Reymen, 2018) adopted to operationalize the innovation ecosystem construct as previously outlined. Following Berglund (2021), design science serves to draw on "rigorous methods to investigate the mechanisms (if any) underlying entrepreneurial practices and to use these insights to propose more useful design knowledge" (p. 3), that is, "practically useful design knowledge and tools" (p. 2). More specifically, the purpose of our study is to develop a *design theory* (Gregor & Hevner, 2013) for mapping, designing and analyzing innovation ecosystems. In doing so, we adopt the anatomy of design theory proposed by Gregor and Jones (2007) to synthesize and instrumentalize the body of knowledge on innovation ecosystems.

This anatomy implies we infer the main constructs and their (causal) relationships from the extant literature, and then craft a visual instantiation of these constructs and their relationships, resulting in a visual representation of EPM for users. The EPM design theory is subsequently applied on a total of 241 (prospective) ecosystem constellations, across seven developmental iterations, which serve to enhance the theory's mutability. We also outline how others have applied this design theory.

Overall, this paper makes a twofold contribution to the literature. First, inspired by Gregor and Jones (2007), we develop a design theory of mapping and evaluating the risk profiles of (prospective) innovation ecosystems, one that covers the entire spectrum from theoretical underpinnings to a body of applications and detailed user guidelines. Second, this study demonstrates the EPM design theory's applicability for early-stage venture teams as well as other types of users. In this respect, EPM design theory provides a generic tool for mapping and analyzing prospective innovation ecosystems.[1]

2. Innovation ecosystems and the need for design theory

Adner (2012, 2017) as well as Williamson and De Meyer (2012) argued that many firms need to develop an innovation ecosystem strategy, which guides the decisions made on their own (ecosystem-oriented) actions as well as their efforts to influence other parties in the ecosystem. Such a strategy should address the structure by which novel value will be created in the ecosystem (Adner & Kapoor, 2010), the roles of external actors within this value structure (Adner, 2017), the (potential) network effects arising from it (Williamson & De Meyer, 2012), how each actor aligns to the value proposition of the ecosystem (Walrave et al., 2018), the influence of each actor's properties on its relative bargaining power and its likelihood to contribute constructively (Adner, 2006; Autio & Thomas, 2014), the risks and value trade-offs of having more (or less) interdependent actors involved in developing the ecosystem (Adner & Feiler, 2019), and the potentially asymmetric interdependence between specific actors and its implications for how they (are likely to) act toward each other (Adner, 2017). We thus assume that key decision-makers on ecosystem strategy have to thoroughly understand what the *specific innovation ecosystem of their new product/service* is like and what the possible implications of any interventions would be.

Generally speaking, practitioners have great difficulties in directly applying extant theories and research results (Radaelli et al., 2014), such as those outlined above. Moreover, at the time we observed that many scholars were struggling with operationalizing the innovation ecosystem concept, with each published article presenting a distinct theoretical framework (e.g. Adner, 2012; Adner & Kapoor, 2010; Autio & Thomas, 2014; Davis, 2016; Iansiti & Levien, 2004; Jacobides et al., 2018; Moore, 1998; Williamson & De Meyer, 2012). Especially Adner (2017) made an important contribution toward synthesizing the extant literature (available at the time), but it remained unclear how an innovation ecosystem could be adequately modeled and represented for both analytical and instrumental-practical purposes.

Around 2014, we set out to develop a comprehensive design theory for mapping and analyzing innovation ecosystems, one that would empower entrepreneurs and other actors to model any envisioned ecosystem and analyze its risk profile in a structured manner. Inspired by Gregor (2006) and Gregor and Jones (2007), we conceived of *design theory* as the deliberate integration of normative and descriptive theories into design paths that are instrumental in producing more effective systems (i.e. innovation ecosystems); the integration of normative and descriptive theorizing into design paths implies that a complete design theory includes its key constructs and relationships between them (cf. descriptive theorizing), various resources and guidelines that support its application (cf. normative theorizing), as well as a body of applications which demonstrates how descriptive and normative theorizing are integrated into design paths.

[1] Late 2018, Long Range Planning published the preprint version of the article presenting EPM (Talmar et al., 2020). In describing how we created and applied EPM design theory in the next sections, we thus stick to its actual timeline in the period 2014-2018. This implies we do not refer to any studies in the innovation ecosystem literature published after that period. The Discussion & Conclusion section does, of course, again draw on more recent work in this area.

As such, a design theory—also called a 'theory for designing'—is distinct from other types of theory commonly observed in academia, that is, theories for analyzing, explaining, and predicting phenomena (Gregor, 2006). That is, a design theory focuses on *how to achieve a desired outcome* (Simon, 1996) by providing a bridge from scholarly work to practical application, which is critical in driving real-world change (Van Aken, 2004).

More specifically, we used the 'anatomy of design theory' proposed by Gregor and Jones (2007). Table 1 outlines this anatomy, as applied and adapted by us to develop a design theory for modeling innovation ecosystems. Notably, the earlier article in LRP (Talmar et al., 2020) merely presented the EPM as a practical method, without describing the complete theory underlying it. The main contribution of the current paper therefore is to report and reflect on how the complete EPM design theory was developed, applied and evaluated.

The first component in Table 1, the design theory's purpose and scope, was elaborated earlier in this section. Section 4 describes the other six components in more detail.

Notably, a complete design theory incorporates all components of the anatomy outlined in Table 1. This differentiates design theory from its more restrictive counterpart in the social sciences, where 'theory' mainly refers to its purpose and scope and a set of core constructs and causal relationships (i.e. the first three components in Table 1). The next section elaborates the methodology of this study in more detail.

Table 1
The anatomy of the design theory for modeling innovation ecosystems (adapted from: Gregor & Jones, 2007).

Component	Description & application
Purpose & scope	*Articulating what the design theory is for.* In the EPM case, the design theory involves a theoretically grounded approach for mapping and analyzing the risk profile of an (prospective) innovation ecosystem, one that empowers key actors to model the envisioned ecosystem of their innovation and analyze its risk profile—that is, the likelihood that this ecosystem can be accomplished.
Constructs	*Representation of constructs as the key entities of interest in the theory.* In EPM design theory, these constructs include the elements and properties which add value to understanding how a real-world (or hypothetical) innovation ecosystem functions in terms of value creation, delivery and capture. As the (prospective) operating logic of any innovation ecosystem depends on both the properties of the actors involved and the properties of the ecosystem network (Dattée et al., 2018), we categorize these constructs accordingly.
Relationships between constructs	*Representation of the links between the constructs as design principles*. For EPM design theory, we develop a set of design principles informed by the extant literature.
Instantiation	*The physical (e.g. graphical) implementation of the theory.* We instantiate the design theory as a graphical artifact that serves as a visual interface for modeling ecosystems.
Mutability	Mutability is defined as "*the changes in state of the artifact anticipated in the theory, that is, what degree of artifact change is encompassed by the theory*" (Gregor & Jones, 2007, p. 322). For EPM design theory, we assess to what extent the key constructs and design principles and their visual instantiation can enable modeling of a diverse set of ecosystems and their stakeholders.
Body of applications	*The body of evidence that demonstrates how the design theory has been applied thus far.* In line with established design science jargon, Gregor and Jones (2007) talk about 'justifications' of the theory. Instead, we refer to 'applications' here, because it is more common in the entrepreneurship and innovation literature. Our own applications of the (preliminary) EPM design theory involve 241 cases. We also report various EPM applications by others.
Guidelines for application	*Detailed articulation of a set of heuristics and processes for applying the design theory, especially its instantiation, to enable others to apply it (independently from the design team that created the theory).* In the case of EPM design theory, this results in a manual with methodological and practical guidelines.

3. Methodology

The underlying methodology of this study is design science (Romme & Reymen, 2018), which is "a problem-solving research strategy that facilitates the production of novel instrumental knowledge for the design and implementation of actions, processes, and systems to achieve desired practical outcomes, that is, to create things that serve human purposes" (Dimov et al., 2023, p. 1543). As described in the previous section, we adopted the anatomy of design theory outlined in Table 1 to synthesize and instrumentalize the extant body of knowledge on innovation ecosystems. The seven key components of this anatomy thereby provide the main steps in the design science cycle of this study (see Fig. 1).

This cycle enabled us to iterate many times through it, in order to gradually move to the final definition of the design theory's purpose and scope, the final set of theoretical constructs and design principles, and the final version of the visual instantiation and the manual with application guidelines. The mutability evaluations and extensive applications in each iteration were instrumental in improving these various components of the design theory. For example, by applying the initial visual instantiation and its underlying constructs to the first few cases (in the 'applying' step), we obtained conceptual and practical insights that helped to improve the definitions of various constructs and design principles as well as the usefulness of the visual instantiation.

Most steps in Fig. 1 involved collecting and/or generating data, in a broad sense. Based on a review of the literature, constructs and their causal relationships were formulated and then synthesized in design principles. Drawing on a review of what kind of archetypical instantiations are available, we chose a visual instantiation and then further developed it into a detailed graphical form. An initial assessment of the theory's mutability was done in the first iteration (after steps 1 to 4) and this assessment was repeated in each subsequent iteration, especially informed by the user experiences and outcomes. The main assessment criterion here was whether the key constructs and design principles and their visual instantiation effectively enable the risk assessment of a diverse set of ecosystems.

In the application step, we used the (preliminary) design theory in various workshops in which the most recent version of the constructs and design principles as well as their visual instantiation were used to map and analyze specific ecosystems. The first version of the (first four components of the) design theory was available early 2015, and we subsequently exposed it to extensive practical tests, also to enhance the mutability of the artifacts it produced.

Table 2 provides an overview of all applications and their learnings. This table also shows that the nature of the applications changed substantially over time: the early efforts focused on improving and sharpening the initial (raw) design theory, also to determine whether it achieves its main purpose (Rescher, 1977); subsequently, the focus shifted toward assessing whether the (increasingly complete) theory can enable substantial mutability in terms of highly different ecosystems.

In applications 1 and 2, we therefore assessed whether participants can apply each construct of the design theory; participants here applied these theoretical components to 122 cases (see Table 2). A substantial number of participants (i.e. 22) also provided survey-based feedback on the various elements of the design theory. These first applications in Table 2 especially served to improve the definition and measurement of ecosystem risk as a core construct.

The design theory was subsequently refined by means of a substantial number of iterations, also reported in Table 2 (i.e. applications 3 to 10), involving another 180 participants. The sixth column in Table 2 outlines the nature and method of evaluation (Hevner et al., 2004) and the last column reports the learnings and results regarding the theory's mutability. At least one author of this paper facilitated each application. The individuals participating in the applications varied from university

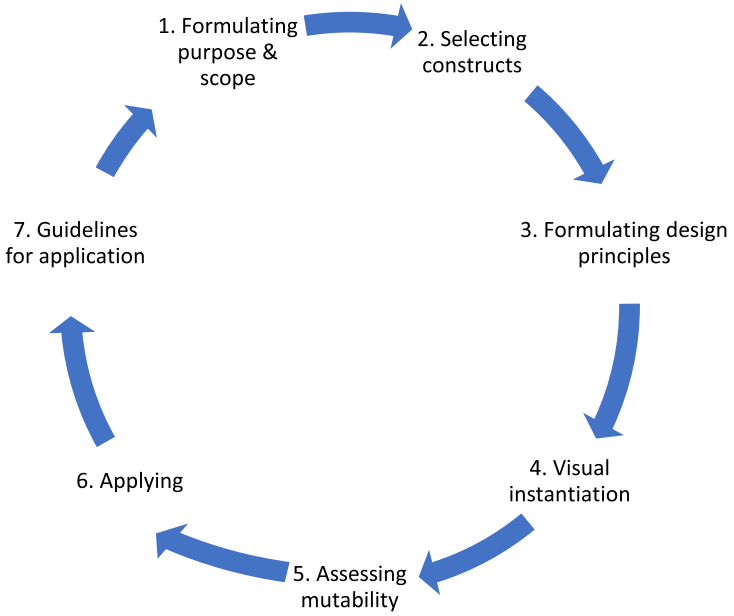


Fig. 1. Design science cycle adopted in this study.

students and scholars to (nascent) entrepreneurs, industry representatives, and representatives from local government, with their weight gradually shifting from the former to the latter as the design theory matured. In total, these application efforts resulted in 241 specific EPM models. We collected the latter templates, as co-created and filled in by participants, from all applications (in their original form and/or as photographs).

4. Creating and applying a design theory for innovation ecosystem modeling

This section describes the key components of the design theory developed. We start with the key constructs (4.1) and their relationships in terms of design principles (4.2), then describe the theory's visual instantiation (4.3), and conduct an initial assessment of its mutability at several artifact levels (4.4). Subsequently, section 4.5 illustrates how the (preliminary components of) EPM design theory was applied, and section 4.6 outlines the guidelines for application. Section 4.7 provides an overview of how others have applied this theory. Finally, section 4.8 reports an in-depth assessment of the theory's mutability, drawing on our own applications as well as applications by others.

4.1. Ecosystem-level and actor-level constructs

Regarding constructs, we distinguished between constructs at the innovation ecosystem level and actor level; see Talmar et al. (2020) for a more detailed overview. The ecosystem-level constructs inferred from the literature are:

- *Ecosystem's value proposition (EVP).* Adner's (2017) structuralist perspective implies an innovation ecosystem is characterized by the search for a system-level goal, in the form of a coherent customer-oriented EVP (Appleyard & Chesbrough, 2017; Clarysse et al., 2014).
- *User segment.* The user segment(s) specifies the target market for the value created in the ecosystem, as formulated in the EVP (Clarysse et al., 2014; Williamson & De Meyer, 2012).
- *Actors.* These are the companies, organizations, institutions and individuals that engage as the main actors in creating, delivering and capturing (future) value in the ecosystem (Adner & Feiler, 2019; Autio & Thomas, 2014), by providing at least one complement (i.e. most commonly a resource, product, service, or decision) required to materialize the EVP. Each actor is legally independent, but within the (envisioned) ecosystem actors are economically (and possibly asymmetrically) interdependent.
- *Structure.* The actors and their value additions (defined below) to the envisioned EVP are in a relational structure to each other.

The actor-level constructs inferred from the literature are:

- *Resources.* Resources are the basis for realizing any actor-specific contribution to the ecosystem (Davis, 2016). The actor may own these resources, but can also obtain them from other actors in the ecosystem (Lubik & Garnsey, 2016), for instance by means of a joint venture or creating shared facilities. Here, resource complementarities are key drivers of value creation (Kapoor & Furr, 2015; Koenig, 2012).
- *Activities.* This construct refers to how an actor is expected to add value to the ecosystem. More specifically, a firm's activity system encompasses how resources are converted into value addition through a process of coordinating within and/or across actor resource interactions (Zott & Amit, 2010). Consequently, similar resources can enable different value additions, depending on the composition of the activities (Hennart, 1994; Zott & Amit, 2010). While 'activities' is an actor-level construct, its boundary-spanning nature implies that distinct activities become connected to deliver value to user segments.
- *Value addition (*also known as *'complement').* As an outcome of its activities, each actor contributes some unique complement to the EVP, for which it is likely to have a comparative advantage in doing so (Autio & Thomas, 2014). The contribution of each actor is its value addition, which can be conceived as a module within the EVP (Clarysse et al., 2014; Nambisan & Sawhney, 2011). Some of these complements may be *critical* for accomplishing the EVP, while others may be *non-critical* (or peripheral) and merely enhance the EVP (Gawer & Cusumano, 2002). This difference is important for designing the actor composition of the ecosystem, but also for assessing its risks (see below).
- *Value capture.* While the ecosystem creates and delivers its end-user value through various interactions, value capture primarily occurs

Table 2
Summary of the applications.

Applications	Participants profile	Number of participants	Purpose of application	Number of models	Method of evaluation	Resulting changes
1. University course, including several modeling workshops	Undergraduate students of diverse areas of study	38	Educate students on the importance of ecosystem thinking in innovation processes. Exercise entailed the modeling of real-life ecosystems (38) across industries, based on archival and interview data; and potential future ecosystems (12) based on complementarities between real-life actors.	50	Outcome assessment + survey + field observations	1. Redevelop argumentation on *Risk*, representing the risk of the actor to the ecosystem, as opposed to risk to the actor. Refine various guidelines, based on survey-based feedback.
2. University course, including several modeling workshops	Undergraduate students of diverse areas of study	55	Educate students on the importance of ecosystem thinking in innovation processes. Exercise entailed the modeling of 55 real-life ecosystems across various industries, based on archival and interview data; and 17 potential future ecosystems based on complementarities between real-life actors.	72	Outcome assessment + survey + field observations	2. Redevelop argumentation on *Flows & Relationships* into *Links*. Rewrite the methodological guidelines.
3. University course, including several modeling workshops	Graduate students of diverse topics of study	39	Educate students on the importance of ecosystem thinking in innovation processes. Course entailed the modeling of new potential ecosystems based on emerging technologies in the domain of urban solutions.	8	Outcome assessment + field observations	3. In the EPM, introduce *color-coding* of Actors to represent the belonging of Actors in different (prospective) value chains.
4. Business development workshop	Venture team in a university, considering the founding of a startup company	18	Generate distinct ecosystem model alternatives for commercializing a particular disruptive technology in the field of energy storage. Articulate the strategic implications of each potential model for the venture team. See the case study in section 4.5.1.	4	Case study + field observations	4. Introduce the construct of *Underlying motivation* of each Actor to participate in the ecosystem. Shift Activities to be embedded in the Value addition construct as the mechanism by which value is created.
5. Conference workshop	Students of industrial engineering	12	Demonstration of ecosystem modeling by designing three potential future ecosystems in smart mobility.	3	Field observations	5. Re-conceptualize *Customer segments* construct to *User segments*
6. Pilot phase of a radically new service ecosystem	Industry, academia	12	Map an existing ecosystem structure in the domain of energy services, analyze it in terms of shortcomings and propose interventions to alleviate these shortcomings.	5	Case study + field observations	6. Shift *User segments* from the core of the ecosystem (coupled with EVP) to be part of *Actors* and thus take active part in value creation and delivery.
7. Developing a new innovation governance process	Industry	6	In an innovation intermediary, develop an innovation governance process for initiating and supporting new technology development ecosystems. See case study reported in section 4.5.2.	Not directly applicable	Case study + field observations	None
8. Two-year research, captured in a comprehensive report	Industry-commissioned engineering researcher	4	Perform a scenario analysis about the possible future ecosystems in the wide-scale commercialization of a generic technological innovation within the electricity sector; develop interventions to accomplish the scenarios.	5	Field observations	None
9. Two business development workshops	Industry, local government, academia	30	Develop business propositions for new products/services, including an explicit ecosystem management element; in the domain of urban solutions.	18	Field observations	7. Remove construct *Underlying motivation*. Introduce the construct of *Dependence*. Shift *Activities* back to being a separate construct.
10. University course, including several modeling workshops	Undergraduate students of diverse areas of study	59	Educate in innovation ecosystem thinking. Exercise entailed the modeling of real-life ecosystems (59) across industries, based on archival and interview data; and potential future ecosystems (17).	76	Outcome assessment + field observations	None

at the level of individual actors (Teece, 1986). The construct of value capture therefore reflects *how* (i.e. the mechanism), *what kind* of, and *how much* value created will be captured by a specific actor. The opportunity to capture value is a key motivation to contribute to the ecosystem (Lepak et al., 2007; West & Wood, 2014) and to sustain commitment, and therefore actors should perceive their 'share' to be sufficient (Iansiti & Levien, 2004; Williamson & De Meyer, 2012). In addition to direct financial gains, an actor may also capture non-monetary value in terms of reputation effects, increased efficiency, or access to additional resources (Lepak et al., 2007), which often implies that non-monetary value is monetized elsewhere.

- *Dependence*. Ecosystems are networks that often include highly different actors (Adner & Kapoor, 2016). For some, accomplishing the EVP may be crucial to their viability and survival; for others, their contribution to the EVP is relatively small compared to their total set of activities (Adner, 2017). An actor's 'dependence' thus refers to the extent to which the (envisioned) EVP's success affects this actor's overall viability and survival chances.
- *Actor-specific risk*. The notion of actor-specific risk is directly related to dependence and value capture and indirectly to all other constructs. For the EVP to materialize, actors have to achieve a sufficient level of agreement, alignment and commitment regarding their contributions (Adner & Silverman, 2013; Koenig, 2012; Williamson & De Meyer, 2012). A key source of risk here is the *unwillingness* of a specific actor to contribute the complement expected, for example, due to a low level of value capture relative to the costs of this actor (Iansiti & Levien, 2004), or the actor's low dependence on the success of the ecosystem, resulting in decreased efforts by this actor (Adner, 2017). Furthermore, actor-specific risk can arise from the (perceived) *inability* of an actor to deliver its value addition (Adner, 2006), for example, due to technological difficulties or lack of resources.

Finally, we can define 'ecosystem risk' by combining the various ecosystem-level and actor-level constructs:

- *Ecosystem risk*. Drawing on conjunctive probability logic (cf. Adner & Feiler, 2019), the likelihood of accomplishing the EVP is determined by multiplying the likelihoods that each critical actor is willing and able to contribute to the EVP. This probabilistic exercise serves to aggregate the actor-specific risks (as an actor-level construct) to a risk assessment of the entire ecosystem, in terms of the likelihood that the EVP can be effectively accomplished within the projected timeline (Adner, 2006).

Ecosystem risk assessment is the primary purpose of ecosystem modeling, as its outcomes can be weighed against expected upsides (e.g. projections of revenue and return-on-investment) to inform the decision about whether or not to develop a specific innovation ecosystem and/or which specific EVP and ecosystem composition is evaluated to be the most promising in terms of its risk/reward ratio.

4.2. Relationships between constructs in terms of design principles

The causal and normative nature of these relationships can be synthesized in four design principles (DPs). The first DP directly follows from the body of work outlined previously (e.g. Adner, 2006; Adner & Feiler, 2019):

DP1. *A set of actors that wants to explore the success chances of an envisioned EVP, for which each actor provides a complement, has to assess the likelihood (or risk) that this innovation ecosystem accomplishes a timely and successful launch of the EVP by (a) estimating the likelihood that each critical complement will be delivered in the required form and at the expected point in time and (b) subsequently multiplying these likelihoods to obtain an overall risk assessment.*

The second design principle outlines what drives the likelihood that a specific actor will deliver its complement (e.g. Adner, 2017; Iansiti & Levien, 2004; Williamson & De Meyer, 2012), that is, it spells out activity (a) in the previous DP:

DP2. *A set of actors that wants to estimate the likelihood that actor X will deliver its critical complement in the required form and at the expected point in time has to assess (a) the willingness as well as (b) the ability of this actor to create and deliver its complement.*

The third and fourth DPs outlines how one can assess and predict an actor's willingness and ability to make the complement available, thereby fleshing out the last part of DP2. Regarding the actor's willingness, the literature refers to incentives in terms of the (expected) ratio of value capture relative to the costs borne by the actor (Iansiti & Levien, 2004), whether or not the actor's resources and activities for the EVP can be redeployed elsewhere (Cennamo et al., 2018), and the actor's dependence on the success of the ecosystem (Adner, 2017). With regard to an actor's (in)ability to make its complement available, Adner (2006) referred to the available human resources, technological capabilities, and legal (incl. intellectual property) capabilities. In sum, the last two DPs are as follows:

DP3. *A set of actors that wants to estimate the likelihood that actor X is willing to deliver its complement in the required form and at the expected point in time has to assess this actor's incentives in terms of the expected value captured, its opportunities to redeploy resources and activities (for the EVP) elsewhere, and its dependence on the success of the ecosystem.*

DP4. *A set of actors that wants to estimate the likelihood that actor X is able to deliver its complement in the required form and at the expected point in time has to assess this actor's available resources, its technological capability and its legal capability.*

This set of DPs allows a focal innovator independently, or the contributors to a nascent ecosystem collectively, to explore whether an EVP and its required ecosystem composition is likely to be accomplished, or alternatively, compare various possible ecosystem compositions in terms of their success chances.

4.3. Instantiation into a visual interface

To instantiate the first two components of the design theory, outlined thus far, in an interface that is user-friendly and visually appealing, we decided to exploit the widely used metaphor of the stakeholder value 'pie' (Tantalo & Priem, 2016). Visually, this 'pie' division of value is commonly represented in the form of a rounded figure, which is divided in sectors that represent particular actors (Bourne & Walker, 2005; Lüftenegger, 2014; Türetken et al., 2019). With the aim of creating a two-dimensional paper-based artifact, we embedded the constructs and propositions previously described in a rounded visual structure. And after several iterations and tryouts, we arrived at the graphical artifact presented in Fig. 2.

Key elements of the EPM artifact are: (1) the ecosystem-level goal of materializing the EVP is placed at the center of the model; (2) the various actors in the ecosystem are separated by radial lines around the center; (3) the clockwise sequence of the actors represents the dominant direction of value transfer in the ecosystem, starting with upstream contributions and finishing with the final user segment(s); (4) the actor-level constructs of resources, activities, value addition, value capture, and risk are mapped for each distinct actor (positioned within its radial lines), and the actor's dependence on the ecosystem's viability and success is assessed on one radial line, using a Low-Medium-High scale; and (5) arrows are used to represent (major) links between constructs, with the direction of the arrow reflecting the direction of value transfer (in case value is transferred in a reciprocal manner, a bi-directional arrow is used).

4.4. Initial mutability assessment

Gregor and Jones (2007) define artifact *mutability* as "the changes in state of the artifact anticipated in the theory, that is, what degree of artifact change is encompassed by the theory" (p. 322). The key constructs and design principles in sections 4.1 and 4.2 constitute the theoretical underpinnings, whereas the visual instantiation in the EPM is the main generic artifact, which in turn enables users to create their own context-specific artifacts (in the form of a detailed risk assessment of their own envisioned or existing ecosystem). At this stage, we therefore have two types of artifacts:

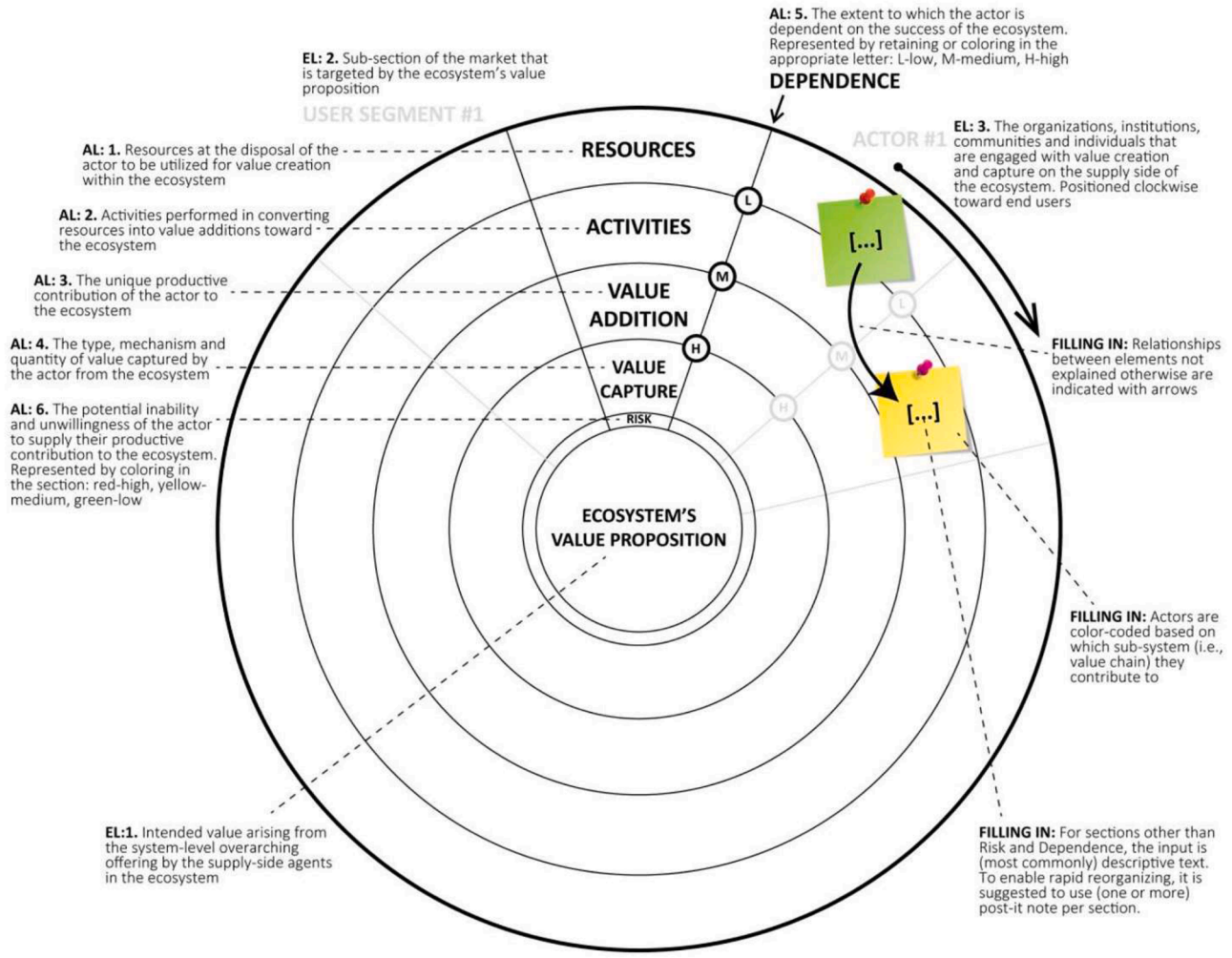


Fig. 2. EPM's visual instantiation as a graphical artifact (source: Talmar et al., 2020, p. 2), with EL referring to Ecosystem Level and AL to Actor Level.

- the generic visualization of the EPM (in Fig. 2); and
- the context-specific EPM graphs filled in by users (to be illustrated in the next section).

At a later stage, we also developed a manual with detailed user guidelines (see section 4.6), which constitutes another essential artifact. However, at this stage of fleshing out the anatomy outlined in Table 1, we only focused on the two artifact levels defined above.

We thus assessed to what extent the key constructs and propositions and their visual instantiation in EPM enable a diverse set of ecosystems to be mapped and analyzed (from an overall risk perspective). This initial mutability assessment was especially done via the first three applications outlined in Table 2, which provided specific feedback from a large number of initial applications of the first versions of EPM design theory. These initial applications were done in the form of modeling workshops with undergraduate and graduate students. The feedback received from these workshops was instrumental in, for example, redeveloping the risk construct, further develop various aspects of the visual instantiation, and fine tune various (initial) user guidelines (see Table 2). Most importantly, these initial applications demonstrated that (student) users could effectively model a wide variety of prospective ecosystems, by filling in the template in Fig. 2 based on publicly available data as well as interviews with key actors. Section 4.8 will report a more final assessment of EPM design theory's mutability.

4.5. *Two applications in more detail*

Section 3 already outlined how the design theory was used and adapted (see applications 4 to 10 in Table 2). In this section, we outline two of these cases in more detail. The first case involves an attempt to commercialize a radically new technology, and the second one is about the development of a new innovation governance approach by a European innovation intermediary.

4.5.1. *Commercializing a radically new technology*

The first case involves case 4 in Table 2. Here, we used the EPM theory in a business development workshop for a venture team that invented a novel solution for stable liquid form hydrogen storage that could power a special kind of electricity generator. The commercialization of this technology would entail a radical shift in the activities performed by several actors in the ecosystem, which might hamper its adoption. Correspondingly, the purpose of this ecosystem modeling exercise was to explore the strategic implications of several alternative applications of the novel technology and to make a well-informed decision on its commercialization strategy. More specifically, the venture team developed and compared four ecosystem alternatives for commercializing the technology: three in the domain of mobility and transport (public transport, trucks and boats) and another one in power storage for buildings.

By modeling each of these ecosystems with EPM design theory, the venture team addressed key questions such as what kind of

complementary products would be needed to materialize the EVP; which actors can create and supply these; which ecosystem alternative is the most likely to become viable, given the risk profiles of the four ecosystem alternatives; how to alleviate the risks specifically associated with each critical actor; which modules of the ecosystem can be internalized by the focal venture and which can be left for others to develop and maintain; what is a suitable mechanism to capture value (e.g. licensing, becoming an OEM) for the venture team, so as to encourage external parties to adopt the technology and accomplish the EVP; what are the implications of that choice for the venture's (e.g. capital) resource requirements; and, finally, which of the alternative ecosystems will the venture team pursue? Fig. 3 shows one of the EPM templates filled in by the venture team.

As result of the various modeling exercises, the team decided to focus on building an ecosystem for its novel technology in public transport (as mapped in Fig. 3). This choice was primarily informed by the huge dependence and risks associated with various actors and their complements in the three alternatives assessed, and the lack of feasible measures to mitigate these risks. Moreover, in the case of these alternative ecosystems in which the power-to-liquid technology is applied to trucks, boats or buildings, it would be difficult to achieve a substantial scale that attracts other major actors. By contrast, in the case of public transport, new policies of municipalities (e.g. regarding requirements for greenhouse gas emissions) are likely to increase the motivation of other key actors to join tenders in the area of public transport. In this respect, the venture team concluded that tender requirements give the public transport operator a pivotal role in tendering for less polluting buses among bus manufacturers, who in turn would need technological solutions to fulfill the requirements. Additionally, while public transport consumes a lot of fuel (providing scale advantages for companies producing and delivering that fuel), it requires far fewer tanking stations than most other mobility alternatives, which in turn greatly increases the likelihood of making the (critically important) fueling infrastructure available. While these traits of the public transport ecosystem option did make the venture team evaluate the risk level of various actors involved as moderate or high (see Fig. 3), it perceived the total risk level of this ecosystem to be considerably lower than those of its three alternatives.

4.5.2. *Developing a new innovation governance approach*

In the second case (i.e. case 7 in Table 2), EPM design theory played a significant role in redesigning the process by which a European innovation intermediary initiates and governs the innovation activities in their industry. One key activity of this intermediary was to fund the development and commercialization of innovation projects accomplished by means of consortia-based innovation. However, its existing approach in this area suffered from three problems.

The first problem arose from allowing applicants to pre-agree on consortia, which often left them 'unbalanced'; for example, organizations that would duplicate each other's efforts, contribute unevenly, or lack the actual incentives and/or ability to integrate the individual work-packages into a new EVP. Second, the intermediary regularly opened calls for program proposals from new consortia, to which many technology firms typically responded. As a result, most accepted proposals were heavily technology-driven, which was considered to be problematic because too little attention was given to business, societal, governmental, and other dimensions. Moreover, with projects and their consortia being built around a specific technological invention, they were often blind to potential complementarities with other (nascent) technologies which would possibly enable a bigger ecosystem and a more valuable EVP. Third, the intermediary was limited to the type of proposals submitted, making it difficult to obtain an overview of various industrial and technological trends and exploit potential synergies

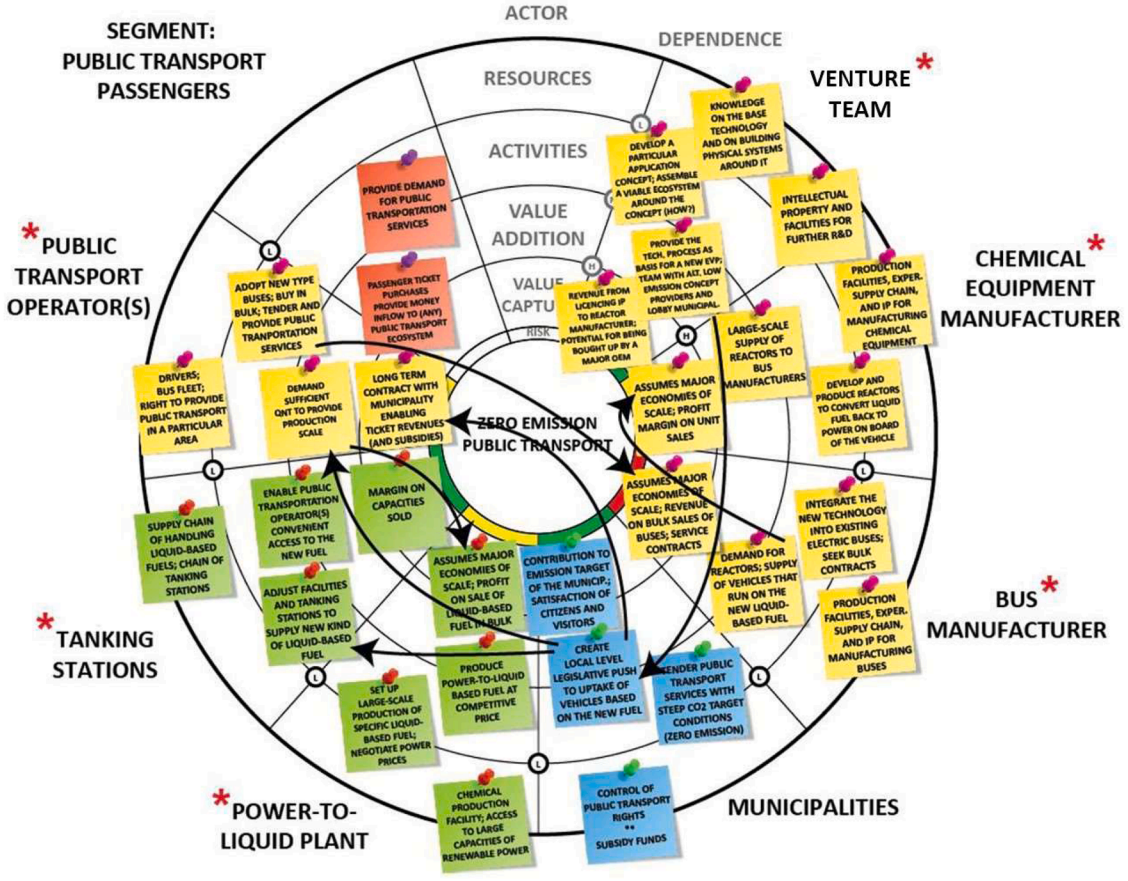


Fig. 3. Ecosystem model for power-to-liquid fuel in public transportation (adapted from: Talmar et al., 2020, p. 7).

between them.

In response to these challenges, the intermediary aimed to develop and implement an alternative process for innovation project evaluation. We helped the intermediary to design this process, by drawing on the EPM design theory's constructs and propositions (see sections 4.1 and 4.2). The newly designed process featured four stages. In the first stage, the intermediary would use the vision articulated in its technology roadmaps and complement it with expert knowledge to identify and prioritize particular bottlenecks that appeared to inhibit the development of a subdomain of the industry. Accordingly, the identified bottlenecks would be understood in a multifaceted way, which in turn served as the basis for a *preliminary* innovation call.

In this preliminary call, the intermediary invited a diverse range of organizations to submit information on (some of the aspects of) the problem addressed. The design of the application form for this preliminary call was informed by the actor-level constructs in section 4.1, asking for (1) what activities the organization already conducts that could contribute to alleviating the problem; (2) what resources (i.e. human and intellectual capital) the organization draws on, in conducting these activities; (3) the potential and drive of collaboration with other organizations; (4) its capabilities and drive toward commercial solutions (as opposed to scientific ones); and (5) a future vision on how to solve the problem, including cues on what kind of ecosystem collaborations are required.

Subsequently, the intermediary considered all applicants submitting proposals to this preliminary call as potential actors in new ecosystems, and thereby started to design prospective complementarities between these actors and envision future innovation ecosystem constellations to address the problem. At this stage, a group of experts at the intermediary used the visual instantiation in Fig. 2 to paper-prototype future ecosystems.

With an initial vision on the structures of several alternative ecosystems to address the complex problem, the intermediary would then invite potentially synergistic actors to consortia development workshops, in which the process of EPM modeling was repeated—but now with the participation of potential actors in these prospective ecosystems. Any promising (nascent) ecosystem emerging from these workshops was then motivated and invited to develop a consortium that applies for innovation project funding, while being continually challenged by the program organizers on whether the consortium was complete and whether all actors were truly willing and able to contribute. This led many (potential) consortia to withdraw from making final funding applications, whereas the final applications submitted were evaluated to be of very high quality.

In this application, EPM design theory served to deliberately redesign the innovation process of a European intermediary, also by fleshing out several key dimensions and tools for the new process. This case is also interesting because EPM design theory was applied by staff members of this intermediary, an organization that facilitates the design and rise of many other innovation ecosystems.

4.6. Developing guidelines for application

The final component of the design anatomy in Table 1 is to develop detailed guidelines for applying EPM design theory. These guidelines should especially also enable other users to apply this theory, independently from the design team that created it. Building on the rich feedback obtained from various applications (as previously reported), the first author of this study wrote an extensive manual with methodological as well as practical guidelines. The initial version of this manual was iteratively adapted and improved, informed by the various applications reported earlier. The most recent version of the manual is available from: https://ecosystempie.com. This manual contains a detailed description of the graphical interface and the modeling process, including:

- what are innovation ecosystems and what is ecosystem strategy;
- the target audiences for EPM;
- the functionality of EPM, also relative to other approaches;
- overview of the building blocks of the EPM;
- examples of ecosystem models;
- frequently asked questions and their answers; and
- various templates to support applications of the EPM.

4.7. Examples of subsequent applications by others

Many others have been applying the EPM design theory, since the first version of the user manual became publicly available in 2019. Some of these applications are outside the scope of the EPM design theory. For example, Helman (2020) used it for assessing the (currently highly fragmented) regional innovation system around the Polish city of Wrocław, to identify promising innovation and entrepreneurial opportunities and develop a regional innovation profile. This type of application raises questions about whether EPM design theory can be effectively applied at a more aggregate (e.g. regional) level than we initially envisioned. In the remainder of this section, we outline four applications that are within its scope, that is, ecosystems in search mode.

De Moraes et al. (2023) examined the key enablers and barriers for alternative protein innovation in Brazil, in view of the environmental, social and animal welfare concerns about animal agriculture and the associated meat industry. They used the EPM design theory's theoretical constructs and their relationships (outlined in sections 4.1 and 4.2) to analyze and synthesize their data from interviews with experts in the field, in order to identify the main enablers and barriers for an ecosystem producing alternative proteins to emerge and thrive.

Peron et al. (2025) used the EPM to design an ecosystem model for additive manufacturing (AM) in the medical sector. They first conducted a systematic literature review to identify AM challenges and opportunities in the medical sector, which were then validated through a Delphi study. The participants in the latter Delphi study were also asked to suggest countermeasures for the challenges and help identify future AM adoption scenarios. Using EPM design theory, these findings were subsequently analyzed and synthesized to design a prospective innovation ecosystem for AM (regarding medical implants and devices) in the medical sector (Peron et al., 2025).

Sourmelis et al. (2024) adopted EPM design theory in a study of the cement industry's search for new cementitious materials that can replace clinker volumes and reduce carbon footprint, in the context of global alumina production that generates approximately 150–170 million tons of bauxite residue (BR), predominately landfilled. Consequently, future collaboration between the alumina and cement industries may provide opportunities to conserve resources and produce low-carbon cements by thermally modifying BR into novel cementitious materials. Sourmelis et al. (2024) therefore evaluated the suitability of different modified BRs for use as cementitious materials and then use the EPM to identify promising business model configurations that would effectively link the alumina and cement industries regarding BR.

Finally, Sylos Labini et al. (2024, p. 69) investigated "the potential of blockchain technology integration into satellite onboard computing systems, with the aim of harnessing smart contracts and improve multi-sided market definitions and machine-to-machine (M2M) relations from an Internet of Things (IoT) perspective." They considered the roles of various actors to design an ecosystem that leverages the benefits of blockchain technology. Here, the EPM was used to represent possible business models and discuss the conditions for their success, including network effects. Sylos Labini et al. (2024, p. 69) concluded that the integration of blockchain technology into satellite computing systems "can enable a new generation of satellite-based IoT applications", which may "offer improved security, privacy, and efficiency, creating significant opportunities for industry actors and end-users."

These various applications further underline the applicability of EPM

design theory, and especially also demonstrate how it supports the *search* for a viable EVP and the ecosystem configuration required to accomplish this proposition.

4.8. An in-depth mutability assessment

Our own applications (in 4.5) and the applications of EPM design theory by others (in 4.7) suggest that EPM design theory is instrumental in mapping the risk profile of highly different ecosystems. We now turn to a more in-depth reflection on the theory's mutability, within its intended purpose and scope (Gregor & Hevner, 2013; March & Smith, 1995).

We started developing EPM design theory in view of the lack of instrumental methods for ecosystem modeling which practitioners can apply. This implies it should inform key actors' decision-making as well as adequately represent the prospective ecosystem. A key criterion for the theory's mutability thus is whether it informs decision-making by insiders as well as for analytical usage by outsiders (e.g. scholars or public policy-makers).

With this criterion in mind, it is clear that the case in section 4.5.1 illustrates how nascent entrepreneurs can use EPM to explore several potential applications of a novel technology in terms of its EVP and ecosystem composition, whereas the case in 4.5.2 illustrates a more complex application in which an innovation intermediary redesigned its innovation approach (and thereby EVP) offered to external parties. Both cases clearly involve insiders applying EPM design theory. By contrast, it is mostly outsiders that have conducted the four applications outlined in section 4.7. That is, the vast majority of the authors of the studies by De Moraes et al. (2023), Peron et al. (2025), Sourmelis et al. (2024) and Sylos Labini et al. (2024) are scholars, with few practitioners involved. Of course, this set of applications of EPM design theory may not adequately represent the actual population of other users (than ourselves), because practitioner-insiders are less likely than scholar-outsiders to publish their results.

Overall, all applications reported in sections 4.5 and 4.7 underline the prospective functionality of EPM design theory. That is, this design theory is especially helpful when a focal actor (possibly together with complementors) seeks to evaluate the risk profiles of the prospective ecosystem(s) needed to accomplish a shared value proposition.

5. Discussion and conclusion

This paper elaborates the broader design theory underlying the Ecosystem Pie Model (Talmar et al., 2020). In doing so, we employed the anatomy of design theory (Gregor & Jones, 2007) to synthesize and instrumentalize the available knowledge on the structuralist approach to innovation ecosystems. That is, after establishing the purpose and scope of the design theory, we inferred the main constructs from the extant literature and synthesized their causal relationships in a set of design principles, then created a visual instantiation of these constructs and their relationships, and subsequently applied this (initial) design theory to a large number of cases; we also developed a manual with user guidelines, explored how others have applied the design theory, and evaluated its mutability. As such, we applied EPM design theory in more than 240 ecosystem constellations, thereby also demonstrating its mutability. Moreover, the previous section also demonstrated that others have been adopting and applying EPM design theory, which underlines its usefulness as an actionable theory.

Overall, this study makes a twofold contribution to literature. First, it develops a design theory of how to map and evaluate the risk profiles of innovation ecosystems, using the anatomy of design theory developed in the field of information systems (Gregor & Jones, 2007). Whereas design theory and its anatomy are well-known in the latter field (Gregor, 2006; Gregor & Hevner, 2013), it is a novel form of theory in the entrepreneurship and innovation discourse, also in terms of how scholars have conceived of design science thus far (e.g. Berglund, 2021; Dimov et al., 2023; Romme & Reymen, 2018; Seckler et al., 2021). In this respect, prior applications of design science by entrepreneurship and innovation scholars focused on creating practical artifacts (Baldassarre et al., 2020; Hyytinen et al., 2023; Romme & Holmström, 2023), at best informed by design principles that detail the underlying theoretical mechanisms (Eckerle & Terzidis, 2024; Pascal et al., 2013; Petzolt & Seckler, 2025; Sagath et al., 2019). Inspired by the discourse on design theory in the adjacent field of information systems, our study may help broaden the notion of descriptive and explanatory theorizing which prevails in the entrepreneurship and innovation (ecosystem) literature. Broadening the notion of theory is essential in bridging the practice-academia gap (Eppler & Platts, 2009; Romme et al., 2015). More specifically, a complete (design) theory not only contains clearly defined constructs and design principles, but also includes various other instantiations (such as a graphical interface and a user manual) that enable its application and help build a body of empirical evidence demonstrating its mutability (Gregor & Hevner, 2013; Gregor & Jones, 2007). This also implies that any set of design principles (e.g. those developed in section 4.2) can only be applied to specific cases *in combination with* these various instantiations. In other words, one has to evaluate a design theory as a whole, because each anatomic component cannot be independently applied to real-world challenges in a meaningful manner.

Second, our study contributes to the literature by demonstrating EPM design theory's applicability for early-stage venture teams, innovation intermediaries, and others prospecting potential applications of novel technologies in terms of their EVPs and ecosystems. This design theory is grounded in the literature (e.g. Adner, 2012; Baldwin et al., 2024; Dattée et al., 2018), but also moves beyond it by providing a generic tool for mapping and analyzing prospective or nascent systems for value co-creation. In this respect, EPM design theory not only appears to capture (prospective or nascent) ecosystems that possibly result in ecosystems characterized by super-modular interdependencies (Jacobides et al., 2018; Sylos Labini et al., 2024) but any exploratory effort to collectively co-create value, including those that subsequently develop into systems that also be effectively conceptualized and analyzed as clusters or supply chains. This broad functionality of EPM theory is comparable to the generic role of scaffolds on construction sites, that is, it provides new venture teams, educators, scholars and other users with "scaffolding structures for creativity and interaction" (Iandoli, 2023, p. 350).

This scaffolding role of EPM design theory empowers scholars studying ecosystems in at least three ways. For one, it enables them to deeply understand the (envisioned) structure of the specific innovation ecosystem at hand (Adner, 2017). Depending on their specific preferences and needs, they can zoom into specific (e.g. dependency and risk) relationships between key actors, or alternatively, use the graphical representation in Fig. 2 to obtain a holistic understanding of the ecosystem as a whole. Users tend to find it difficult to estimate the implications of interdependencies for an innovation ecosystem's success (Adner & Feiler, 2019), and EPM design theory might help them assess these implications in a realistic manner and devise strategies accordingly. Second, when one applies EPM design theory to represent and analyze different ecosystems, especially the visual instantiation in Fig. 2 enables systematic comparison of research output, which is critical in building a cumulative body of knowledge on innovation ecosystems. Thirdly, EPM design theory appears to generate substantial insights in prospective ecosystems (e.g. Helman, 2020; Sourmelis et al., 2024; Sylos Labini et al., 2024), implying that ecosystem scholars can offer (to supervise) the application of this theory in exchange for access to data collection.

In terms of the maturity scale for design theories developed by Gregor and Hevner (2013), EPM theory currently qualifies as a *nascent* design theory. A nascent design theory goes beyond a situated implementation of an artifact, but requires further testing and refining to become a well-developed design theory. In turn, the nascent nature of EPM design theory calls for future work that further develops its core

constructs, graphical instantiation, user manual, and so forth. This may also help in building a comprehensive body of knowledge on innovation ecosystems, one that draws on a single set of well-defined constructs, a common set of artifacts, and a strong body of empirical evidence.

A limitation of this study is that EPM design theory was developed from 2014 to 2018, resulting in Talmar et al.'s (2020) presentation, which was primarily written for an audience of practitioners at the time. Advancements in the ecosystem literature as of 2018 are, therefore, not yet incorporated in this theory. Future work in this area should therefore explore whether recent insights in, for example, ecosystem complementarities and externalities (Jacobides et al., 2024) and user and producer pathways to complementarity (Borner et al., 2023) can be incorporated in EPM design theory or, alternatively, motivate the development of other design theories. In a recent conceptual paper, Bröring and Thybussek (2023) extended EPM theory with other constructs such as technology-product-market linkages, technology system, and technology innovation system—which obviously calls for studies applying this extended framework. In any case, future studies in this area should, ideally, draw on experimental approaches that generate more robust evidence regarding the performance and benefits of these (EPM or any other) design theories.

In conclusion, innovation ecosystems have become pivotal phenomena in a world of increasing specialization and complexity, one in which single ventures often cannot accomplish value propositions on their own. In this paper, we described and reflected upon how EPM design theory was developed and also demonstrated its applicability.

CRediT authorship contribution statement

Madis Talmar: Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Jan Holmström:** Writing – review & editing, Supervision, Methodology, Conceptualization. **A. Georges L. Romme:** Writing – review & editing, Supervision, Funding acquisition. **Bob Walrave:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Ksenia Podoynitsyna:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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